



Aishu R 251001007-2025-IT ▾

A2**Started on** Tuesday, 4 August 2026, 9:01 AM**State** Finished**Completed on** Tuesday, 4 August 2026, 9:03 AM**Time taken** 1 min 55 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  void reverse(int n)
3  {
4      int rev = 0, remainder;
5      int count=0;
6      count=count+1;
7      while (n != 0)
8      {
9          count=count+2;
10         remainder = n % 10;
11         count=count+1;
12         rev = rev * 10 + remainder;
13         count=count+1;
14         n /= 10;
15         //count=count+1;
16     }
17     count=count+2;
18     printf("%d",count);
19     //print(rev);
20 }
21 int main(){
22     int n;
23     scanf("%d",&n);
24     reverse(n);
25     return 0;
26 }
```

	Input	Expected	Got	
✓	12	11	11	✓

	Input	Expected	Got	
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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